

FLEXHAND · DESIGN DECISIONS REGISTER

FlexHand design decisions register

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	FXH-DEC-001	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/flexhand		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Mass savings tried (11 g off the pack, 723 to 712 g); cost USD 304.90; R7 still not met
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations for all five open decisions on 2026-10-02 (FXH-DDR-003 accepted); moved to decisions made; value engineering states that the mass savings cannot close the R7 gap
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the review note, FXH-DDR-001 to FXH-DDR-003 and the build plan work

FlexHand design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The gearbox face's two M3 holes: spacing (17 mm assumed), how deep a screw may go, and the shaft length (12.5 mm assumed)	They set the bulkhead holes, the 6 mm screw length and how far the spool hub reaches onto the shaft	FXH-DDR-003, C3, C4
2	The motor length with its encoder (71 mm assumed) and where the encoder cable leaves	Sets the 6 mm cable gap behind the motors	FXH-DDR-003, C5
3	The stop button's body depth (18 mm assumed), bezel and nut	Its body sits between the cells and the motors	FXH-DDR-003, C5
4	The sizes of the controller, driver, charger, battery board and regulator modules, and the charger's USB-C port position	They are laid out on the tray, and the port must meet the rear wall opening	FXH-DDR-003, C5
5	The cells' length with any protection fitted (65 mm assumed)	The saddles and tray posts leave 0.5 mm at each end	FXH-DDR-003, C5
6	The sheath housing's outside diameter (4 mm) and its ferrules' (about 5 mm)	They set the puck, stop block and cover holes	FXH-DDR-003, C7, C11
7	The glove's size and stretch against the plate and spacer positions	Plates and spacer are stitched where the model puts them	FXH-DDR-003, C11, C12
8	The breakaway couplings' pull-out force (about 60 N) and the stop-bead settings	Set and recorded at TRL 4	FXH-DDR-002, N3, N4

■ Value engineering

Value-engineering target: USD 500 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 304.90 (USD 195.10 under the target), from `bom/bom.csv` [J1]. Main cost drivers and savings worth trying:

- **Gearmotors, USD 107.90 (36 %).** The largest line and 214 g of the pack's mass. Worth pricing a lighter 20 to 25 mm gearmotor class with encoder that still carries 0.49 N·m peak; any change reopens R2 and the torque calculation.
- **Battery, stop button and electronic modules, about USD 68.** Standard modules; little to save in cost, but a single dual-channel driver board would save one board, its wiring and a few grams.

- **Anchor block and its hardware, USD 18 plus fixings.** The printed parts are now 68 g, 3.4 g less after a larger lightening pocket (tried on 2026-10-02). A thinner-walled block with the channels as separate tubes, or printing it in a lighter material, remains to try.
- **Fixing kit, USD 11, 22 g.** Aluminium M3 screws now hold the lid and the cover (tried on 2026-10-02): 5.4 g saved for USD 1 more. The tray keeps steel self-tapping screws because aluminium ones are not made in that size.
- **Pack base, 103 g.** Saddle ribs thinned from 4 to 3 mm saved 2.6 g (tried on 2026-10-02). Lower infill is a print setting that saves little in 2 mm walls and is left to the TRL 4 prints; the motor torque reaction is checked there.

The savings tried on 2026-10-02 (decision 5) took 11.4 g off the pack, from 723 g to 712 g, and cannot close the 262 g gap to R7's 450 g target. R7 stays not met, and the therapist review of the target, which also treats the pack's load on a weak arm and shoulder as a safety question, decides it.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D8: 25 mm 227:1 gearmotors on the forearm, two motors, passive motion only, widest anchor cuffs with a soft liner, 30 N design load, passive thumb, tendons in Bowden sheaths, 2S pack	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	FXH-DDR-001
2026-09-25	TRL 4 on hold for this repo	Amish, same instruction	<code>project.yaml</code> (trl_target), FXH-DDR-001
2026-09-25	Items N1 to N5: R2 restated for mild tone, fingertip thimbles, balance pulleys with stop beads, ball-detent couplings and a perforated cuff, 4 s lower stroke limit at design load	Amish: "i accept all your recommendations, go with them across all repos."	FXH-DDR-002
2026-09-26	FlexHand in the first batch of product renders	Amish chose the repo (no words recorded)	<code>docs/REVIEW.md</code> , session of 2026-09-26
2026-09-30	Make the design physically buildable while drawing the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	FXH-DDR-003 (changes made under this instruction; accepted on 2026-10-02, below)
2026-09-30	Outstanding decisions are kept in this register, not in the build plan	Amish: "don't log outstanding decisions in this build plan - that is not the place for it. that should be in a separate design document logged and named as such"	This register
2026-10-01	<code>budget_usd</code> is a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	This register, Value engineering
2026-10-02	First clinical co-design partner: a university rehabilitation lab attached to a stroke service, introduced through the OpenRatio network; first candidate to approach: Shirley Ryan AbilityLab in Chicago, or a university rehabilitation program nearer Irving	Amish: "i approve your recommendations for all 555 open decisions."	FXH-DDR-001, 01
2026-10-02	Design for construction accepted: changes C1 to C14 and their knock-on changes, as made; the release (C7) and the finger spread (C10) are confirmed separately in the next two rows	Amish: "i approve your recommendations for all 555 open decisions."	FXH-DDR-003, Tables 1 and 2

Date	Decision	Decided by	Record
2026-10-02	Tool-free release: the pull-out plate (option a). Pass mark set before the TRL 4 test: released in 10 s or less (R9) with a pull force that the co-design therapist agrees, before the test, a care partner can apply with one hand; the cam lever (option b) is the fallback if it fails	Amish: "i approve your recommendations for all 555 open decisions."	FXH-DDR-003, A1
2026-10-02	Finger spread: option (a), cuffs that hold the fingers slightly spread, but nothing is worn until the co-design therapist has checked the spread on a range of hands with mild tone; switch to single-saddle cuffs if the spread raises tone or discomfort	Amish: "i approve your recommendations for all 555 open decisions."	FXH-DDR-003, A2
2026-10-02	Forearm pack about 723 g: keep the 450 g target of R7 under therapist review, treating the pack as a safety question for that review (load on a weak arm and shoulder), and try the listed mass savings before TRL 4	Amish: "i approve your recommendations for all 555 open decisions."	FXH-DDR-003, A3; FXH-DDR-002, N4